StateFarm Distracted Driver Detection Model

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Abstract

Approached with the task of detecting distracted drivers, we studied image-based distracted driver detection models on the StateFarm Distracted Driver Detection dataset, viewing several images from the dashboard including: safe driving, texting, phone calls, and reaching behind the passenger seat. Distracted driving is a leading cause of road accidents, and automated in-cabin monitoring could enable timely driver alerts. We trained and fine tuned models from Kaggle's competition using [key choices: augmentation, transfer learning, loss], and we evaluated with a driver-disjoint 70-30 test split. Our best model achieves an accuracy of [X%]. Denote about errors These results suggest

1. Literary Analysis

To find a basis for creating a model, we first started with a literary analysis to find effective methods and shortcomings. Each participant read numberous articles and summarized the teams' results.

1.1. Distracted Driving Detection using CNNs

Four CNN models were used to training their[1] collected data using transfer learning and softmax regression: VGG-16 model, AlexNet, GoogleNet, and ResNet. The best performance came from the ResNet model, performing an accuracy of 92%, out performing the other models by up to 6%. The researchers concluded that most of the incorrect classifications from the models were due to the similarities in postures, such as drinking, hair and makeup, and texting on the phone with the left hand. One possible implementation to help reduce the number of misclassifications is to combine their overall approach with a face-based approach or to replace the final fully connected layers with classifiers such as SVM or HMM.

1.2. Distracted Driving Monitoring with Artificial Intelligence

The article[3], written by Hui Xiong and Zhiming Zhang (2026). The authors note in their overall abstract for the paper, that one of the leading causes of traffic accidents occur due to distracted driving. In the final experiment, the total F1 score used to measure performance of the model is represented by comparing the correct predictions to the actuality of the dangerous driving. In the five types of driving tested, normal, speeding, hard-breaking, distracted, and drowsy driving, all F1 scores exceed 90% with the exception of hard braking receiving an F1 score of 88.48%. The system has an overall high performance, especially with consideration to the abnormal driving behaviors. Xiong and Zhang were able to implement a system to test for distracted driving in the form of a behavioral monitoring system through the use of machine learning and computer vision techniques alongside artificial intelligence. The biggest short coming for their final model is more complex scenarios and overall road safety.

1.3. Paper length

Papers, excluding the references section, must be no longer than eight pages in length. The references section will not be included in the page count, and there is no limit on the length of the references section. For example, a paper of eight pages with two pages of references would have a total length of 10 pages. **There will be no extra page charges for 5031 2026.**

Overlength papers will simply not be reviewed. This includes papers where margins and formatting are deemed to have been significantly altered from those laid down by this style guide. Note that this LaTeX guide already sets the figure captions and references in a smaller font. The reason why such papers will not be reviewed is that there is no provision for supervised revisions of manuscripts. The review process cannot determine the suitability of the paper for presentation in eight pages if it is reviewed in 11 pages.

1.4. The ruler

The LaTeX style defines a printed ruler that should be present in the version submitted for review. The ruler is provided in order that reviewers may comment on particular lines in the paper without circumlocution. If you are preparing a document using a non-LaTeX document preparation system, arrange for an equivalent ruler to appear on the final output pages. The presence or absence of the ruler should not change the appearance of any other content on the page. The camera-ready copy should not contain a ruler. (LaTeX users may use the options of `cvpr.sty` to switch between different versions.)

Reviewers: note that the ruler measurements do not align well with lines in the paper — this turns out to be very difficult to do well when the paper contains many figures and equations, and, when done, looks ugly. Use fractional references (*e.g.*, this line is 087.5), although in most cases the approximate location would be adequate.

1.5. Paper ID

Make sure that the Paper ID from the submission system is visible in the version submitted for review (replacing the "*****" you see in this document). If you are using the LaTeX template, **make sure to update paper ID in the appropriate place in the tex file**.

1.6. Mathematics

Please, number all of your sections and displayed equations as in these examples:

$$E = m \cdot c^2 \tag{1}$$

and

$$v = a \cdot t. \tag{2}$$

It is important for the reader to be able to refer to any particular equation. Just because you did not refer to it in the text does not mean that some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like "the equation second from the top of page 3 column 1". (Note that the ruler will not be present in the final copy, so is not an alternative to equation numbers). All authors will benefit from reading Mermin's description of how to write mathematics: http://www.pamitc.org/documents/mermin.pdf.

1.7. Blind review

Many authors misunderstand the concept of anonymizing for blind review. Blind review does not mean that one must remove citations to one's own work—in fact it is often impossible to review a paper unless the previous citations are known and available.

Blind review means that you do not use the words "my" or "our" when citing previous work. That is all. (But see below for tech reports.)

Saying "this builds on the work of Lucy Smith [1]" does not mean that you are Lucy Smith; it says that you are building on her work. If you are Smith and Jones, do not say "as we show in [7]", say "as Smith and Jones show in [7]" and at the end of the paper, include reference 7 as you would any other cited work.

An example of a bad paper just asking to be rejected:

> An analysis of the frobnicatable foo filter.
>
> In this paper, we present a performance analysis of our previous paper [1], and show that it is inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.
>
> [1] Removed for blind review

An example of an acceptable paper:

> An analysis of the frobnicatable foo filter.
>
> In this paper, we present a performance analysis of the paper of Smith *et al*. [1], and show it to be inferior to all previously known methods. Why the previous paper was accepted without this analysis is beyond me.
>
> [1] Smith, L and Jones, C. "The frobnicatable foo filter, a fundamental contribution to human knowledge". Nature 381(12), 1-213.

If you are making a submission to another conference at the same time that covers similar or overlapping material, you may need to refer to that submission to explain the differences, just as you would if you had previously published related work. In such cases, include the anonymized parallel submission [**?**] as supplemental material and cite it as

> [1] Authors. "The frobnicatable foo filter", F&G 2014 Submission ID 324, Supplied as supplemental material `fg324.pdf`.

Finally, you may feel you need to tell the reader that more details can be found elsewhere and refer them to a technical report. For conference submissions, the paper must stand on its own, and not *require* the reviewer to go to a tech report for further details. Thus, you may say in the body of the paper "further details may be found in [**?**]". Then submit the tech report as supplemental material. Again, do not assume that the reviewers will read this material.

Sometimes your paper is about a problem that you tested using a tool that is widely known to be restricted to a single institution. For example, let's say it's 1969, you have solved a key problem on the Apollo lander, and you believe that the 1970 audience would like to hear about your solution.

The work is a development of your celebrated 1968 paper entitled "Zero-g frobnication: How being the only people in the world with access to the Apollo lander source code makes us a wow at parties", by Zeus *et al*.

You can handle this paper like any other. Do not write "We show how to improve our previous work [Anonymous, 1968]. This time we tested the algorithm on a lunar lander [name of lander removed for blind review]". That would be silly, and would immediately identify the authors. Instead write the following:

> We describe a system for zero-g frobnication. This system is new because it handles the following cases: A, B. Previous systems [Zeus et al. 1968] did not handle case B properly. Ours handles it by including a foo term in the bar integral.
>
> ...
>
> The proposed system was integrated with the Apollo lunar lander, and went all the way to the moon, don't you know. It displayed the following behaviours, which show how well we solved cases A and B: ...

As you can see, the above text follows standard scientific convention, reads better than the first version, and does not explicitly name you as the authors. A reviewer might think that it is likely that the new article was written by Zeus *et al*., but cannot make any decision based on that guess. He or she would have to be sure that no other authors could have been contracted to solve problem B.

FAQ

Q: Are acknowledgements OK?
A: No. Leave them for the final copy.

Q: How do I cite my results reported in open challenges?
A: To conform with the double-blind review policy, you can report results of other challenge participants together with your results in your paper. However, for your results, you should not identify yourself and should not mention your participation in the challenge. Instead, present your results referring to the method proposed in your paper and draw conclusions based on the experimental comparison with other results.

1.8. Miscellaneous

Compare the following:

`$conf_a$`	$conf_a$
`$\mathit{conf}_a$`	conf_a

See The TEXbook, p165.

The space after *e.g*., meaning "for example", should not be a sentence-ending space. So *e.g*. is correct, *e.g.* is not. The provided `\eg` macro takes care of this.

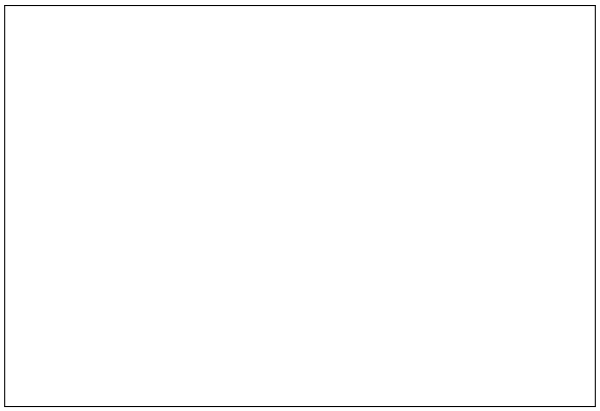


Figure 1. Example of caption. It is set in Roman so that mathematics (always set in Roman: $B \sin A = A \sin B$) may be included without an ugly clash.

When citing a multi-author paper, you may save space by using "et alia", shortened to "*et al*." (not "*et. al.*" as "*et*" is a complete word). If you use the `\etal` macro provided, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al*. However, use it only when there are three or more authors. Thus, the following is correct: "Frobnication has been trendy lately. It was introduced by Alpher [**?**], and subsequently developed by Alpher and Fotheringham-Smythe [**?**], and Alpher *et al*. [**?**]."

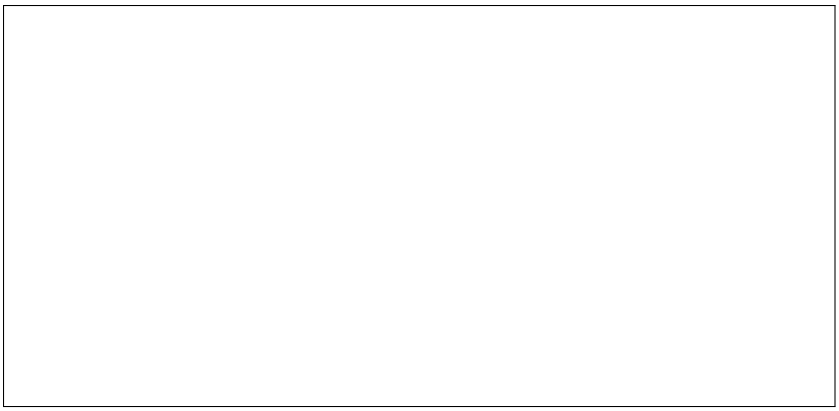
This is incorrect: "... subsequently developed by Alpher *et al*. [**?**] ..." because reference [**?**] has only two authors.

2. Formatting your paper

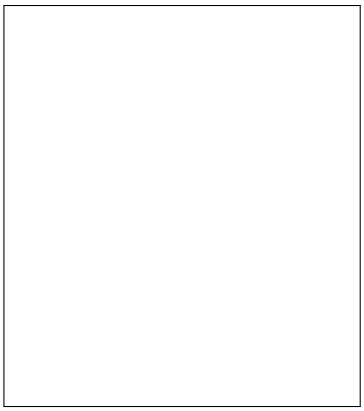
All text must be in two-column format. The total allowable size of the text area is $6\frac{7}{8}$ inches (17.46 cm) wide by $8\frac{7}{8}$ inches (22.54 cm) high. The columns should be $3\frac{1}{4}$ inches (8.25 cm) wide, with a $\frac{5}{16}$ inch (0.8 cm) space between them. The main title (on the first page) should begin 1 inch (2.54 cm) from the top edge of the page. The second and following pages should begin 1 inch (2.54 cm) from the top edge. On all pages, the bottom margin should be $1\frac{1}{8}$ inches (2.86 cm) from the bottom edge of the page for 8.5×11-inch paper; for A4 paper, approximately $1\frac{5}{8}$ inches (4.13 cm) from the bottom edge of the page.

2.1. Margins and page numbering

All printed material, including text, illustrations, and charts, must be kept within a print area $6\frac{7}{8}$ inches (17.46 cm) wide by $8\frac{7}{8}$ inches (22.54 cm) high. Page numbers should be in the footer, centered, and $\frac{3}{4}$ inches from the bottom of the page. The review version should have page numbers, yet the final version submitted as camera ready should not show any page numbers. The LATEX template takes care of this when used properly.



(a) An example of a subfigure.



(b) Another example of a subfigure.

Figure 2. Example of a short caption, which should be centered.

2.2. Type style and fonts

Wherever Times is specified, Times Roman may also be used. If neither is available on your word processor, please use the font closest in appearance to Times to which you have access.

MAIN TITLE. Center the title $1\frac{3}{8}$ inches (3.49 cm) from the top edge of the first page. The title should be in Times 14-point, boldface type. Capitalize the first letter of nouns, pronouns, verbs, adjectives, and adverbs; do not capitalize articles, coordinate conjunctions, or prepositions (unless the title begins with such a word). Leave two blank lines after the title.

AUTHOR NAME(s) and AFFILIATION(s) are to be centered beneath the title and printed in Times 12-point, non-boldface type. This information is to be followed by two blank lines.

The ABSTRACT and MAIN TEXT are to be in a two-column format.

MAIN TEXT. Type main text in 10-point Times, single-spaced. Do NOT use double-spacing. All paragraphs should be indented 1 pica (approx. $\frac{1}{6}$ inch or 0.422 cm). Make sure your text is fully justified—that is, flush left and flush right. Please do not place any additional blank lines between paragraphs.

The captions of the figures and tables should be in 9-point Roman type as in Figs. 1 and 2. Short captions should be centered. Table captions should be above tables, while figure captions should be below figures.
Callouts should be 9-point Helvetica, non-boldface type. Initially capitalize only the first word of section titles and first-, second-, and third-order headings.

FIRST-ORDER HEADINGS. (For example, **1. Introduction**) should be Times 12-point boldface, initially capitalized, flush left, with one blank line before and one blank line after.

SECOND-ORDER HEADINGS. (For example, **1.1. Database elements**) should be Times 11-point boldface, initially capitalized, flush left, with one blank line before and one after. If you require a third-order heading (we discourage it), use 10-point Times, boldface, initially capitalized, flush left, preceded by one blank line, followed by a period, and your text on the same line.

2.3. Footnotes

Please use the footnotes[1] sparingly. Indeed, try to avoid footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence). If you wish to use a footnote, place it at the bottom of the column on the page on which it is referenced. Use Times 8-point type, single-spaced.

2.4. Cross-references

For the benefit of author(s) and readers, please use the

```
\cref{...}
```

command for cross-referencing to figures, tables, equations, or sections. This will automatically insert the appropriate label alongside the cross-reference as in this example:

> To see how our method outperforms previous work, see Fig. 1 and Tab. 1. It is also possible to refer to multiple targets as once, *e.g.* to Figs. 1 and 2a. You may also return to Sec. 2 or look at Eq. (2).

If you do not wish to abbreviate the label, for example, at the beginning of the sentence, you can use

```
\Cref{...}
```

command. Here is an example:

> Figure 1 is also quite important.

[1] This is what a footnote looks like. It often distracts the reader from the main flow of the argument.

Table 1. Results. Ours is better.

Method	Frobnability
Theirs	Frumpy
Yours	Frobbly
Ours	Makes one's heart Frob

2.5. References

List and number all bibliographical references in 9-point Times, single-spaced, at the end of your paper. When referenced in the text, include the citation number in square brackets, for example [**?**]. Where appropriate, include page numbers and the name(s) of editors of referenced books. When citing multiple papers at once, make sure that you cite them in numerical order such as this [**? ? ? ? ?**]. If you use the template as advised, this will be taken care of automatically.

2.6. Illustrations, graphs, and photographs

All graphics should be centered. In LaTeX, avoid using the `center` environment for this purpose, as this adds potentially unwanted whitespace. Instead, use

```
\centering
```

at the beginning of your figure. Please ensure that any point you wish to make is resolvable in a printed copy of the paper. Resize fonts in figures to match the font in the body text, and choose line widths that render effectively in print. Readers (and reviewers), even of an electronic copy, may choose to print your paper in order to read it. You cannot insist that they do otherwise and, therefore, must not assume that they can zoom in to see tiny details on a graphic.

When placing figures in LaTeX, it is almost always best to use `\includegraphics`, and to specify the figure width as a multiple of the line width as in the example below

```
\usepackage{graphicx} ...
\includegraphics[width=0.8\linewidth]
                {myfile.pdf}
```

2.7. Color

Please refer to the author guidelines on the 5031 2026 web page for a discussion of the use of color in your document.

If you use color in your plots, please keep in mind that a significant subset of reviewers and readers may have a color vision deficiency; red-green blindness is the most frequent kind. Hence, avoid relying only on color as the discriminative feature in plots (such as red *vs*. green lines), but add a second discriminative feature to ease disambiguation.

3. Final copy

You must include your signed IEEE copyright release form when you submit your finished paper. We MUST have this form before your paper can be published in the proceedings.

Please direct any questions to the production editor in charge of these proceedings at the IEEE Computer Society Press: https://www.computer.org/about/contact.

References

[1] Yunsheng Ma, Runjia Du, Amr Abdelraouf, Kyungtae Han, Rohit Gupta, and Ziran Wang. Driver digital twin for online recognition of distracted driving behaviors. *IEEE TPAMI*, 9, 2024. Face and Gesture submission ID 324. Supplied as supplemental material `fg324.pdf`. 1

[2] Duy Tran, Ha Mahn Do, Weihua Sheng, He Bai, and Girish Chowdhary. Real-time detection of distracted driving based on deep learning. *IET*, 12, 2018.

[3] Hui Xiong and Zhiming Zhang. Design and implementation of a driving behavior monitoring system based on artificall intelligence. *IJAASR*, 5, 2026. Supplied as supplemental material `tr.pdf`. 1